

Cellular Networks

GSM, GPRS & EDGE:

1. **What is the idea behind GSM? What is its access scheme? Why is FDD preferred over TDD? Describe the mixture of FDMA and TDMA used for GSM.**

Idea: GSM is a digital mobile communication system designed to provide voice and data services over a cellular network. It aims to enable seamless communication between mobile devices across different geographic locations.

Access scheme: GSM uses a combination of Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA). FDMA divides the frequency spectrum into multiple channels, while TDMA divides each channel into time slots for multiple users.

Preference for FDD over TDD: Frequency Division Duplexing (FDD) is preferred over Time Division Duplexing (TDD) in GSM because FDD provides separate frequency bands for uplink and downlink communication, minimizing interference and improving signal quality. TDD shares the same frequency band for both uplink and downlink, which can lead to increased interference and reduced performance.

2. **GSM handover: what's the idea? What's the difference to a handover in a WLAN system?**

Idea: GSM handover involves transferring an ongoing call or data session from one cell to another as a mobile device moves through the network. The goal is to maintain seamless communication and ensure optimal signal strength and quality.

Difference to WLAN handover: In GSM, handover decisions are primarily based on signal strength and quality measurements between neighboring cells, while WLAN handovers may also consider factors like network load and channel conditions.

3. **Is there a handover if a GSM mobile does not have an active call? Why or why not?**

No handover occurs for inactive GSM mobiles (i.e., not engaged in an active call) because handovers are typically initiated to maintain call continuity and optimize signal quality for active calls. Inactive mobiles are not actively communicating and do not require handover to maintain connectivity.

4. **How does authentication work in GSM? Who or what is authenticated? How is the integrity of the shared key ensured?**

GSM authentication verifies the identity of the mobile device and the network before allowing access to network services. The mobile device and the network authenticate each

other using a shared secret key called the Authentication Key (Ki). The integrity of the shared key is ensured through secure key distribution mechanisms and encryption protocols.

5. What is the difference between hard and soft handovers?

Hard handover: Involves momentarily breaking the connection with one cell before establishing a connection with another cell. It is more suitable for scenarios where there is a significant difference in signal strength between cells.

Soft handover: Allows a mobile device to simultaneously connect to multiple cells during the handover process, gradually transitioning the connection from one cell to another. It is more suitable for scenarios with overlapping coverage areas and smoother handover transitions.

6. What is the difference between make-after-break and make-before-break?

Make-after-break: Involves establishing a connection with the target cell after disconnecting from the source cell. It is the traditional approach to handover.

Make-before-break: Involves establishing a connection with the target cell before disconnecting from the source cell, ensuring seamless handover without interrupting communication.

7. What are the main differences between a GSM and a WLAN system? Compare the two systems with respect to media access, radio resource management and handover execution.

Media access: GSM uses TDMA/FDMA for channel access, while WLAN typically uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).

Radio resource management: GSM relies on centralized network control for resource allocation and handover decisions, while WLAN systems may use distributed or centralized approaches.

Handover execution: GSM handovers are primarily based on signal strength measurements, while WLAN handovers may consider factors like network load and interference.

8. Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate) for GSM voice and GPRS/EDGE.

Reuse of 3: Provides higher capacity and data rates compared to a reuse of 9 due to reduced interference and improved signal quality.

Quality (error rate): A reuse of 3 may offer better error rates than a reuse of 9 due to lower interference levels and improved signal-to-noise ratios.

However, the exact impact on capacity and quality may vary depending on specific network configurations, traffic patterns, and deployment scenarios for GSM voice and GPRS/EDGE.

UMTS (Universal Mobile Telecommunication System)

1. Which multiplexing scheme is employed by UMTS?

UMTS (Universal Mobile Telecommunications System) employs a combination of Code Division Multiple Access (CDMA) and Time Division Duplexing (TDD) for multiplexing. CDMA allows multiple users to share the same frequency band by assigning unique spreading codes to each user, while TDD divides the transmission time into alternating uplink and downlink slots.

2. What does macro diversity mean?

Macro diversity in UMTS refers to the use of multiple antennas at different base stations (nodes) to improve signal reception and diversity gain. By combining signals received from multiple antennas, macro diversity enhances communication reliability, reduces fading effects, and improves overall system performance, especially in areas with challenging propagation conditions.

3. What does cell breathing in UMTS mean?

Cell breathing in UMTS refers to the dynamic adjustment of cell coverage area and cell capacity based on traffic load and user demand. When the traffic load decreases, the cell's coverage area expands (breathes out) to accommodate more users and maintain service quality. Conversely, when the traffic load increases, the cell's coverage area shrinks (breathes in) to maintain efficient resource utilization and prevent congestion.

4. How does mobility management work in UMTS?

Mobility management in UMTS involves tracking the location of mobile devices as they move within the network and ensuring seamless handover between different cells to maintain ongoing communication sessions. It includes functions such as location registration, handover initiation, and context transfer between serving and target cells. UMTS mobility management protocols ensure continuous connectivity and QoS support for mobile users, even during handover events.

5. How does QoS management work in UMTS?

QoS (Quality of Service) management in UMTS involves prioritizing and guaranteeing the quality of communication services based on user requirements and network capabilities. It includes mechanisms for traffic classification, admission control, and traffic shaping to allocate network resources efficiently and meet service-level agreements (SLAs). UMTS QoS management ensures that different types of traffic, such as voice, video, and data, receive appropriate levels of performance, latency, and reliability based on their QoS parameters.

LTE (Long-Term Evolution)

1. What is the advantage of a common IP-based packet core over separated networks for packet-switched and circuit-switched traffic?

The advantage of a common IP-based packet core in LTE (Long-Term Evolution) is the convergence of packet-switched and circuit-switched traffic onto a single network infrastructure. This convergence simplifies network architecture, reduces operational costs, and enhances resource utilization. It allows for efficient handling of diverse traffic types, seamless mobility, and easier implementation of new services and applications.

2. What is the purpose of opportunistic scheduling? What is the disadvantage?

Opportunistic scheduling in wireless communication systems aims to maximize spectral efficiency and system capacity by dynamically allocating resources to users with favorable channel conditions. The disadvantage of opportunistic scheduling is that it may lead to unfairness or starvation for users with poor channel conditions, as they may receive fewer resources compared to users with better channel conditions.

3. How does interference cancellation work? What is the idea?

Interference cancellation is a signal processing technique used in wireless communication systems to mitigate the effects of interference. The idea is to estimate and remove interfering signals from received signals before demodulation and decoding. Interference cancellation works by using advanced algorithms and multiple antennas to separate desired signals from interfering signals, improving signal quality and system performance.

4. What is the goal of self-organization in LTE?

The goal of self-organization in LTE is to automate the configuration, optimization, and management of network resources and parameters without human intervention. Self-organization enables LTE networks to adapt dynamically to changing environmental conditions, traffic patterns, and user demands, improving network performance, reliability, and efficiency.

5. What are the use cases for self-organized network management in LTE?

Use Cases for Self-Organized Network Management in LTE:

Automatic network configuration: Self-organized network management in LTE can automatically configure network parameters such as cell parameters, neighbor relations, and radio resource allocation.

Self-healing: LTE networks can automatically detect and recover from network failures or performance degradation by dynamically adjusting parameters and reallocating resources.

Load balancing: Self-organization in LTE enables dynamic load balancing across cells and sectors to optimize resource utilization and ensure uniform coverage and capacity distribution.

Interference management: LTE networks can autonomously mitigate interference sources, optimize frequency reuse, and adapt transmission parameters to minimize interference and improve signal quality.